CSPB Course Reviews

Project Demo

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Agenda

- Purpose & User Story
- Our Project & MVP
- Core Features
- High-level Architecture
- Our Tools & Process
- Our Team
- Demo
- Future Features & Improvements
- Team Dynamics, Challenges, & Reflection
- Questions

Purpose & User Story

Problem:

For years, CSPB students have a hard time finding course reviews, which are scattered and buried in various places (Buff Portal course evals, Discord, Reddit, blogs, messages)

Needs:

Find, browse, and write course reviews more easily in one place

Solution:

A website built to help students review CSPB courses

Our Project & MVP

CSPB Course Reviews

A website that enables CSPB students to centrally browse, write, and manage course reviews

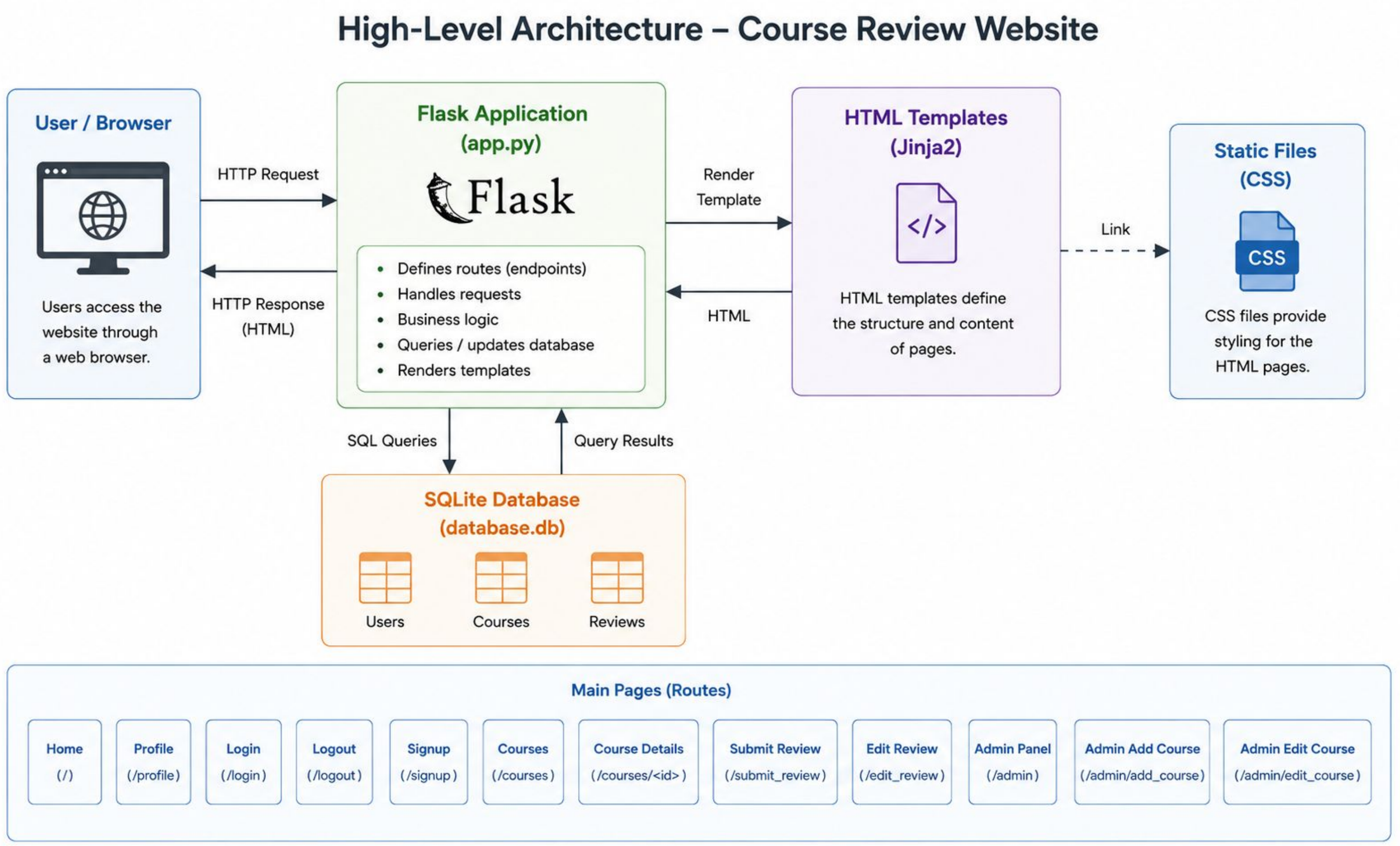
MVP

Our MVP delivers essential features, all tested and functional

Core Features

CSPB Course Reviews

Feature	Guest (Not logged in)	User (Logged in)	Admin (Administrator)
Create an account Sign up for a new account	●	●	●
Log in / Log out Access or end your session	●	●	●
Browse courses & reviews View all CSPB courses and their reviews	●	●	●
Write one review per course Share your experience with a course		●	●
Edit your reviews Update your existing reviews		●	●
Upvote / Downvote / Flag reviews Engage with and report reviews		●	●
Admin Panel (Admin only) Create/edit courses, view users, and moderate flagged reviews			●


High-Level Architecture – Course Review Website
User / Browser
Users access the website through a web browser.
HTTP Request
HTTP Response (HTML)
Flask Application (app.py)
Flask
Defines routes (endpoints)
Handles requests
Business logic
Queries / updates database
Renders templates
Render Template
HTML
HTML Templates (Jinja2)
</>
HTML templates define the structure and content of pages.
Link
Static Files (CSS)
CSS
CSS files provide styling for the HTML pages.
SQL Queries
Query Results
SQLite Database (database.db)
Users
Courses
Reviews
Main Pages (Routes)
Home (/)
Profile (/profile)
Login (/login)
Logout (/logout)
Signup (/signup)
Courses (/courses)
Course Details (/courses/<id>)
Submit Review (/submit_review)
Edit Review (/edit_review)
Admin Panel (/admin)
Admin Add Course (/admin/add_course)
Admin Edit Course (/admin/edit_course)

Our Tools

Tool & Use	Evaluation
Google Docs: Brainstorm, manage tasks & milestones, organize & share data	**Excellent**: Ended up being our main planning & design document. Commenting feature useful for asynchronous work
Discord: Discuss designs, make decisions, and coordinate tasks	**Excellent**: Most used for daily communication and informal check-in meetings
Zoom: Hold regular team meetings	**Good**: Used for formal meetings and milestone recordings
GitHub: Manage code and send pull requests	**Excellent**: Invaluable for version control
Trello: Manage tasks and have task visuals	**OK**: After using Google Docs, was redundant and mostly went unused

Our Process

1. Propose project ideas
2. Clarify user stories, software behaviors, and define the MVP scope
3. Create mockups to illustrate product design
4. Create Flask routes and HTML templates
5. Design and create SQLite database and API methods
6. Connect the front-end and the back-end
7. Test the prototype, troubleshoot issues, and add missing pieces

Our Team

- **All**
  - Flask routes, SQL, CSS, and HTML related to our individual pages
- **Hannah**
  - Main Flask backend structure
  - Overall website CSS styling, HTML skeletons for each page
  - Pages: Base, Home, About, Browse Courses, Course Details
- **Adam**
  - Flask user sessions and authentication
  - Pages: Login, Sign Up, Logout, Profile, User Update Info
- **Sean**
  - SQL backend structure and methods (dbAPI, create and reset db)
  - Pages: Admin Panel, Admin Edit & Add Course, Edit Review
- **Craig**
  - Repo structure / skeleton
  - models.py backend methods and course objects
  - Pages: Submit Review form and related functions for inserting into db

Demo

CSPB Course Reviews

Home Browse Profile Log Out

Welcome back, Algorithm_Puppy!

Share your experience and help other students make informed decisions by writing a review for your courses.

Browse Courses

Recent Reviews

Spring 2026

CSPB 1300: Computer Science 1: Starting Computing

Reviewed by Database_Hawk

Reviewed on 2026-06-23

Overall Rating: **5 / 5** Difficulty: **3 / 5** Workload: **6 hrs /wk**

The final project was fun. I learned a lot.

View Course

Spring 2024

Future Features & Improvements

- Profile/course pictures
- Username, email, & password validation
- Expand admin operations
- More review features (commenting)
- Search filters for courses & reviews
- Better responsiveness for smaller screens
- Increase database resilience

Reflection - Team Dynamics

- **New team members**
  - Different work styles and experience levels while learning and implementing software development methodologies
  - **Mitigation:** A lot of communication, divide work out to play to individual strengths, & help each other out when needed.
- **Time management**
  - Our actual development time was less than one month. As a result, our last weeks were very intense.
  - In retrospect, we could have finished the design phase a little earlier and started coding a little sooner.

Reflection - Team Dynamics

- **Collaborative Learning / Problem Solving**
    - Although our team members had different skill sets and experience levels, everyone was willing to help each other whenever someone had trouble understanding a topic from the course. Working together made it easier to solve problems and learn from one another.

Reflection - Technical Challenges

- **Technical dependencies and different time zones**
  - There were several technical dependencies (ex: back-end, DB, and methods had to be finished to implement front-end features), waiting to merge PRs, etc.
  - **Mitigation:** Weekly team meeting to regroup and daily communication on Discord
- **Great learning experience**
  - Despite unfamiliarity with tools, we managed to deliver all core features.
  - This project has taught us how to build a software product, from a mere idea to a functional and scalable prototype.

Questions?

Thank You!